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Inventor(s)	Lee; Bang Woo

Ceiling panel mounting support structure with improved mounting structure

Abstract

A ceiling panel mounting support structure with an improved mounting structure includes: an E-bracket **130** installed below a slab of a building, a turnbuckle **160** coupled to the E-bracket **130**, and a ceiling panel hanger **180** including a pair of mounting portions **182** on both sides based on a center and coupled to the turnbuckle **160**, wherein ceiling panels **120** arranged to be adjacent to each other are mounted on the pair of mounting portion **182** of the ceiling panel hanger **180**.

Inventors:	Lee; Bang Woo (Cheongju-si, KR)
Applicant:	Lee; Bang Woo (Cheongju-si, KR)
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Primary Examiner: Mintz; Rodney

Attorney, Agent or Firm: Studebaker Brackett PLLC

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims the benefit of Korean Patent Application No. 10-2024-0025442, filed on Feb. 22, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

1. Field

(2) The present disclosure relates to a ceiling panel mounting support structure with improved mounting structure, and more particularly, to a ceiling panel mounting support structure capable of stably installing a ceiling panel on a bottom surface of a building slab, etc.

2. Description of Related Art

(3) In general, pipes, ducts, and electrical wirings are installed in a branched manner below a slab, and ceiling panels are installed to prevent these from being exposed to the outside and to make them look aesthetically pleasing. That is, a plurality of ceiling panels are installed below the ceiling slab to finish the lower portion of the ceiling slab of a building, and items necessary for the building, such as lighting, are installed on the ceiling panels.

(4) Here, a ceiling panel support structure that may stably install a plurality of ceiling panels by mounting them below the ceiling slab is required, and furthermore, a ceiling panel support structure with a further improved mounting structure is required.

RELATED ART DOCUMENT

Patent Document

(5) Korean Patent No. 10-0834861 (registered on May 28, 2008) Korean Patent No. 10-1613697 (registered on Apr. 12, 2016) Korean Patent No. 10-0383793 (registered on Apr. 29, 2003) Korean Utility Model No. 20-0186256 (registered on Apr. 7, 2000)

SUMMARY

(6) An aspect of the present disclosure may provide a ceiling panel mounting support structure capable of stably installing a ceiling panel on a lower surface of a building slab, etc.

(7) According to an aspect of the present disclosure, a ceiling panel mounting support structure with an improved mounting structure includes: an E-bracket **130** installed below a slab of a building, a turnbuckle **160** coupled to the E-bracket **130**, and a ceiling panel hanger **180** including a pair of mounting portions **182** on both sides based on a center and coupled to the turnbuckle **160**, wherein ceiling panels **120** arranged to be adjacent to each other are mounted on the pair of mounting portion **182** of the ceiling panel hanger **180**.

(8) The E-bracket **130** may include a vertical connecting bracket piece **132**, an upper bracket piece **134** with a base portion coupled to one surface of the connecting bracket piece **132**, a middle bracket piece **136**, and a lower bracket piece **138** and having an E-shaped cross-section, the upper bracket piece **134** of the E-bracket **130** may be disposed on an upper surface of a lower beam piece **116** of an H beam **110** installed below the building slab, the middle bracket piece **136** may be fixed by an upper fixing bolt **136UFB** so that the E-bracket **130** is fixedly installed to the H beam **110**, a long bolt **140** may be coupled to the lower bracket piece **138** of the E-bracket **130**, and an upper connecting nut **146** and a lower connecting nut **148** coupled to a threaded portion of an outer circumferential surface of the long bolt **140** may be respectively arranged on upper and lower surfaces of the lower bracket piece **138** of the E-bracket **130** so that the long bolt **140** is fixed to the lower bracket piece **138** of the E-bracket **130**, the turnbuckle **160** may have a ring shape with a pair of left and right side buckle pieces **166** on an upper buckle piece **162** and a lower buckle piece **164**, and the threaded portion of the outer circumferential surface of the long bolt **140** may be coupled to a buckle-side bolt **172** hole formed in the upper buckle piece **162** so that the turnbuckle **160** is coupled to the long bolt **140**, an I-bolt **170** may be coupled to the lower buckle piece **164** of the turnbuckle **160**, and the ceiling panel hanger **180** may be coupled to the I-bolt **170** by a bolt **172** and a nut **174**.

(9) The long bolt **140** may include an upper long bolt **142** having an upper end coupled to the lower bracket piece **138** of the E-bracket **130** and a lower end to which a pipe hanger **150** is coupled and a lower long bolt **144** coupled to a lower end of the pipe hanger **150**, and an upper connecting nut **146** and a lower connecting nut **148** fastened to a threaded portion of an outer circumferential surface of the lower long bolt **144** may be respectively arranged on upper and lower surfaces of the lower bracket piece **138** of the E-bracket **130** so that the long bolt **140** is fixed to the lower bracket piece **138** of the E-bracket **130**, the buckle-side bolt **172** hole formed in the upper buckle piece **162** of the turnbuckle **160** may be coupled to a threaded portion of an outer circumferential surface of the lower long bolt **144** so that the turnbuckle **160** is coupled to the lower long bolt **144**, and the ceiling panels **120** may be installed by mounting the ceiling panels **120** adjacent to each other on a pair of the mounting portions **182** of the ceiling panel hanger **180**.

(10) The ceiling panel mounting support structure of the present disclosure has the effect of stably installing a plurality of ceiling panels by mounting them below a ceiling slab and has the effect of further improving the mounting structure of the ceiling panel. The present disclosure has the effect of reliably providing a ceiling panel mounting support structure with further improved mounting structure and stability, which enables stably installing a plurality of ceiling panels by mounting them below a ceiling slab.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** is a front view schematically illustrating a state in which a ceiling panel is supported and installed by a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure;

(3) FIG. **2** is a front view schematically illustrating a state in which a ceiling panel is supported and installed by a ceiling panel mounting support structure with an improved mounting structure according to another embodiment of the present disclosure;

(4) FIG. **3** is a side view of an I-bolt, which is a major part, illustrated in FIGS. **1** and **2**;

(5) FIG. **4** is a perspective view illustrating a state in which a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure is installed on an H beam below a building slab;

(6) FIG. **5** is a front view illustrating an enlarged state in which a ceiling panel is installed by mounting the ceiling panel by a ceiling panel hanger illustrated in FIG. **2**;

(7) FIG. **6** is a side view schematically illustrating a modified embodiment of a ceiling panel hanger, which is a major portion of a ceiling panel mounting support structure with an improved mounting structure and an enlarged state in which a ceiling panel is mounted according to the present disclosure;

(8) FIG. **7** is a front view schematically illustrating a state in which a ceiling panel is installed and supported by a ceiling panel mounting support structure with an improved mounting structure and a state in which a connecting reinforcement bracket is rotated downward according to another embodiment of the present disclosure;

(9) FIG. **8** is a front view illustrating a state in which the connecting reinforcement bracket illustrated in FIG. **7** is coupled to an H beam; and

(10) FIG. **9** is a front view illustrating an embodiment in which a seismic connecting support is coupled to a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure.

DETAILED DESCRIPTION

(11) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The objects, features, and advantages of the present disclosure will be more easily understood by referring to the accompanying drawings and the following detailed description. In describing exemplary embodiments of the present disclosure, a detailed description of known techniques associated with the present disclosure unnecessarily obscures the gist of the present disclosure, it is determined that the detailed description thereof will be omitted.

(12) In describing elements of the present disclosure, the terms “first,” “second,” “A,” “B,” “(a),” and “(b)” may be used. These terms are merely for differentiating one element from another element, and the essence, number, order or sequence of a corresponding component should not be limited by the terms. Also, when an element or layer is described as being “connected” or “coupled” to another element or layer, the element or layer may be not only directly connected or coupled to that other element or layer, but may also be indirectly connected or coupled to the other element or layer with one or more intervening elements or layers therebetween.

(13) Specific structures and functions in embodiments of the present disclosure are exemplified to illustrate embodiments according to the spirit of the present disclosure, and the embodiments according to the spirit of the present disclosure may be implemented in various forms and should not be construed as being limited to the embodiments described in this specification or application.

(14) FIG. 1 is a front view schematically illustrating a state in which a ceiling panel is supported and installed by a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure, FIG. 2 is a front view schematically illustrating a state in which a ceiling panel is supported and installed by a ceiling panel mounting support structure with an improved mounting structure according to another embodiment of the present disclosure, FIG. 3 is a side view of an I-bolt, which is a major part, illustrated in FIGS. 1 and 2, FIG. 4 is a perspective view illustrating a state in which a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure is installed on an H beam below a building slab, FIG. 5 is a front view illustrating an enlarged state in which a ceiling panel is installed by mounting the ceiling panel by a ceiling panel hanger illustrated in FIG. 2, and FIG. 6 is a side view schematically illustrating a modified embodiment of a ceiling panel hanger, which is a major portion of a ceiling panel mounting support structure with an improved mounting structure and an enlarged state in which a ceiling panel is mounted according to the present disclosure.

(15) Referring to the drawings, the ceiling panel mounting support structure with an improved mounting structure according to the present disclosure includes an E-bracket **130**, a turnbuckle **160**, and a ceiling panel hanger **180** as basic components.

(16) Meanwhile, in another embodiment of the present disclosure, a long bolt **140**, which is a main portion, includes an upper long bolt **142** and a lower long bolt **144**, a pipe hanger **150** is coupled to the upper long bolt **142**, the lower long bolt **144** is coupled to a lower end portion of the pipe hanger **150**, the turnbuckle **160** is coupled to a lower end portion of the lower long bolt **144**, and the ceiling panel hanger **180** may be coupled to the turnbuckle **160** via an I-bolt **170**.

(17) The E-bracket **130** is installed below a slab of a building.

(18) The turnbuckle **160** is coupled to the E-bracket **130**. The turnbuckle **160** is configured in a ring shape with an upper buckle piece **162** and a lower buckle piece **164** provided at upper and lower end portions of the left and right buckle bars, respectively.

(19) The ceiling panel hanger **180** is coupled to the turnbuckle **160**. The ceiling panel hanger **180** is provided with a pair of mounting portions **182** on both sides based on the center. The ceiling panel hanger **180** is coupled to the turnbuckle **160** and is disposed in a vertical direction, and the pair of left and right mounting portions **182** are arranged in a horizontal direction.

(20) In the present disclosure, the ceiling panels **120** arranged to be adjacent to each other are mounted on the pair of mounting portions **182** of the ceiling panel hanger **180**.

(21) In the present disclosure, the E-bracket **130** is fixedly installed on a lower beam piece **116** of

an H beam **110** placed below a slab of a building, so that the E-bracket **130** is installed below the building slab.

(22) The E-bracket **130** includes a vertical connecting bracket piece **132**, an upper bracket piece **134** with a base portion coupled to one surface of the connecting bracket piece **132**, a middle bracket piece **136**, and a lower bracket piece **138** and having an E-shaped cross-section.

(23) The upper bracket piece **134** of the E-bracket **130** is disposed on an upper surface of a lower beam piece **116** of an H beam **110** installed below the building slab, the middle bracket piece **136** is fixed by an upper fixing bolt **136UFB** so that the E-bracket **130** is fixedly installed to the H beam **110**.

(24) The upper bracket piece **134** of the E-bracket **130** contacts an upper surface of the lower beam piece **116** of the H beam **110**, an upper fixing bolt **136UFB** is coupled to a bolt **172** hole penetrating upper and lower surfaces of the middle bracket piece **136** of the E-bracket, an upper end portion of the upper fixing bolt **136 UFB** is mounted above the middle bracket piece **136** so that the upper fixing bolt **136UFB** is pressed on a lower surface of the lower beam piece **116** of the H-beam, thereby stably fixing the E-bracket **130** to the H-beam **110** disposed below the building slab. The connecting bracket piece **132** of the E-bracket **130** is disposed in the vertical direction, and the upper bracket piece **134**, the middle bracket piece **136**, and the lower bracket piece **138** of the E-bracket **130** are arranged in a horizontal direction.

(25) In addition, a long bolt **140** is coupled to the lower bracket piece **138** of the E-bracket **130**, and an upper connecting nut **146** and a lower connecting nut **148** coupled to a threaded portion of an outer circumferential surface of the long bolt **140** are respectively arranged on upper and lower surfaces of the lower bracket piece **138** of the E-bracket **130** so that the long bolt **140** is fixed to the lower bracket piece **138** of the E-bracket **130**. In this manner, the long bolt **140** is fixed to the E-bracket **130** and disposed in the vertical direction.

(26) The turnbuckle **160** has a ring shape with a pair of left and right side buckle pieces **166** on an upper buckle piece **162** and a lower buckle piece **164**, and the threaded portion of the outer circumferential surface of the long bolt **140** is coupled to a buckle-side bolt **172** hole formed in the upper buckle piece **162** so that the turnbuckle **160** is coupled to the long bolt **140**.

(27) In addition, an I-bolt **170** is coupled to the lower buckle piece **164** of the turnbuckle **160**. The I-bolt **172** is configured in a shape having a circular bolt-coupling ring at a lower end portion thereof (See FIG. 3).

(28) The ceiling panel hanger **180** is coupled to the I-bolt **170** by the bolt **172** and a nut **174**.

(29) The ceiling panel hanger **180** has a shape having a hook for coupling a hanger-side bolt **172** that penetrates through both sides of an upper end portion thereof, and by inserting the bolt **172** into the hook for coupling the hanger-side bolt **172** of the ceiling panel hanger **180** and the hook for coupling the bolt **172** of the I-bolt **172** and fastening the nut **174** to the bolt **172**, the ceiling panel hanger **180** is coupled to the I-bolt **170** by the bolt **172** and the nut **174**.

(30) In an embodiment of the present disclosure, the E-bracket **130**, the long bolt **140**, the turnbuckle **160**, the I-bolt **172**, and the ceiling panel hanger **180** are installed sequentially in the vertical direction from top to bottom below the building slab, and the ceiling panels **120** adjacent to each other are mounted on the pair of left and right mounting portions **182** of the ceiling panel hanger **180**. Preferably, caulking is performed between a bottom surface of the ceiling panel **120** and the pair of left and right mounting portions **182** of the ceiling panel hanger **180**.

(31) Therefore, the ceiling panel **120** support structure of the present disclosure has the effect of stably installing a plurality of ceiling panels by mounting them below the ceiling slab and has the effect of further improving the mounting structure of the ceiling panel **120**. The present disclosure has the effect of reliably providing the ceiling panel **120** support structure that may be stably installed by mounting a plurality of ceiling panels below the ceiling slab.

(32) In particular, by fixing the E-bracket **130** to the H beam **110** placed below the building slab and installing the ceiling panel hanger **180** via the E-bracket **130**, a base portion that supports the

ceiling panel hanger **180** that supports the ceiling panel **120** mounted thereon is configured as the more solid E-bracket **130**, so that the effect of reliably providing the ceiling panel **120** support structure capable of stably installing a plurality of ceiling panels by mounting them may be expected.

(33) Here, as illustrated in FIG. 6, the ceiling panel hanger **180** is further provided with a pair of upper ceiling panel support pieces **183** on the left and right so as to be arranged above the pair of left and right mounting portions **182**, so that the pair of left and right mounting portions **182** support the ceiling panel **120** from below, and the upper ceiling panel support piece **183** supports the ceiling panel **120** from above, thereby enabling the ceiling panel **120** to be installed more firmly and stably. In FIG. 5, it is illustrated that the hanger-side hook portion provided on the ceiling panel hanger **180** is installed to be caught by a hook portion provided in a lower end portion of the I-bolt **170**.

(34) Meanwhile, in another embodiment of the present disclosure, the long bolt **140** includes an upper long bolt **142** having an upper end coupled to the lower bracket piece **138** of the E-bracket **130** and a lower end to which a pipe hanger **150** is coupled and a lower long bolt **144** coupled to a lower end of the pipe hanger **150**, and an upper connecting nut **146** and a lower connecting nut **148** fastened to a threaded portion of an outer circumferential surface of the lower long bolt **144** are respectively arranged on upper and lower surfaces of the lower bracket piece **138** of the E-bracket **130** so that the long bolt **140** is fixed to the lower bracket piece **138** of the E-bracket **130**, the buckle-side bolt **172** hole formed in the upper buckle piece **162** of the turnbuckle **160** is coupled to a threaded portion of an outer circumferential surface of the lower long bolt **144** so that the turnbuckle **160** is coupled to the lower long bolt **144**.

(35) In another embodiment of the present disclosure, the E-bracket **130**, the upper long bolt **142**, the pipe hanger **150**, the lower long bolt **144**, the turnbuckle **160**, the I-bolt **172**, and the ceiling panel hanger **180** are sequentially installed in the vertical direction from top to bottom below the building slab, and the ceiling panels **120** adjacent to each other are mounted and installed on the pair of left and right mounting portions **182** of the ceiling panel hanger **180**.

(36) In another embodiment of the present disclosure, as described above, in the present disclosure, the ceiling panels **120** adjacent to each other are mounted on the pair of mounting portions **182** of the two ceiling panel hangers **180** arranged to be adjacent to each other so as to be installed below the building slab. Preferably, in another embodiment of the present disclosure, caulking is performed between the bottom surface of the ceiling panel **120** and the pair of left and right mounting portions **182** of the ceiling panel hanger **180**.

(37) In another embodiment of the present disclosure, by inserting and mounting the pipe **152** in the pipe hanger **150**, it is possible to install more pipes **152** for a building required for building construction below the building slab, and it is also possible to expect the effect of stably installing the building pipe **152**.

(38) Meanwhile, FIG. 7 is a front view schematically illustrating a state in which a ceiling panel is installed and supported by a ceiling panel mounting support structure with an improved mounting structure and a state in which a connecting reinforcement bracket is rotated downward according to another embodiment of the present disclosure, FIG. 8 is a front view illustrating a state in which the connecting reinforcement bracket illustrated in FIG. 7 is coupled to an H beam, and FIG. 9 is a front view illustrating an embodiment in which a seismic connecting support is coupled to a ceiling panel mounting support structure with an improved mounting structure according to the present disclosure.

(39) The present disclosure further includes a connecting reinforcement bracket **190**, and the connecting reinforcement bracket **190** includes a reinforcement connecting piece **192** connected to the lower bracket piece **138** of the E-bracket **130** via a hinge HG, a cross reinforcement connecting piece **193** connected in a direction perpendicular to the reinforcement connecting piece **192**, an upper reinforcement connecting piece **194** connected to an inner surface of the cross reinforcement

connecting piece **193** and disposed in a direction parallel to the reinforcement connecting reinforcement connecting piece **196** connected to the inner surface of the cross reinforcement connecting piece **193** so as to be disposed below the upper reinforcement connecting piece **194** and extending in a direction parallel to the upper reinforcement connecting piece **194**, and a reinforcement connecting support bolt **198** coupled to the lower reinforcement connecting piece **196** and having an upper end portion that may protrude from one side of the lower reinforcement connecting piece **196** to the other side.

(40) The H beam **110** described in the present disclosure includes an upper beam piece **114** and a lower beam piece **116** at upper and lower end portions of a web **112**, respectively. In a state in which the connecting reinforcement bracket **190** is rotated downward based on the hinge at the E-bracket **130** as illustrated in FIG. 7, the connecting reinforcement bracket **190** is rotated upward based on the hinge at the E-bracket **130** as illustrated in FIG. 8. Then, the reinforcement connecting support bolt **198** is tightened in the bolt **172** coupling hole that penetrates the upper and lower surfaces of the cross reinforcement connecting piece **193** of the connecting reinforcement bracket **190** so that the upper end portion of the reinforcement connecting support bolt **198** is pressed to be in close contact with a lower surface of the lower beam piece **116** of the H beam **110**. As a result, the connecting reinforcement bracket **190** is stably coupled to the H beam **110** located below the building slab and maintained.

(41) Therefore, in the present disclosure, since the connecting reinforcement bracket **190** is separately fixed to the H-beam **110** on the opposite side of the E-bracket **130** while the E-bracket **130** is fixed to the H-beam **110** below the building slab, the E-bracket **130** may be reliably prevented from escaping from the H-beam, and since the E-bracket **130** is reliably prevented from escaping from the H-beam, the ceiling panel hanger **180** and the ceiling panel **120** may be reliably and stably installed below the building slab.

(42) Meanwhile, as described above, the H beam **110** described in the present disclosure has the upper beam piece **114** and the lower beam piece **116** at the upper and lower end portions of the web **112**, respectively. In the present disclosure, a reinforcement support block **202** is interposed between the upper reinforcement connecting piece **194** of the connecting reinforcement bracket **190** and an upper surface of the lower beam piece **116** of the H beam **110**, and a block fixing reinforcement bolt **204**, which is fastened to the upper reinforcement connecting piece **194** and presses, fixes the reinforcement support block **202** to the upper surface of the lower beam piece **116** of the H beam **110** is further provided.

(43) By further providing this block fixing reinforcement bolt **204**, the connecting reinforcement bracket **190** may be stably fixed to the H beam **110** below the building slab without shaking.

(44) In addition, in the present disclosure, the upper long bolt **142** of the long bolt **140** is divided into a first upper long bolt **142A** and a second upper long bolt **142B**, and an earthquake-resistant connecting support **210** connecting the first upper long bolt **142A** and the second upper long bolt **142B** is further provided. The earthquake-resistant connecting support **210** includes an earthquake-resistant connecting sleeve **212** in which an upper bolt **172** insertion hole at an upper end portion is slidably coupled to an outer circumferential surface of the first upper long bolt **142A** and a lower bolt **172** insertion hole at a lower end portion is slidably coupled to an outer circumferential surface of the second upper long bolt **142B**, a first upper hook flange **213** provided at a lower end portion of the first upper long bolt **142A** and hooked on an inner surface of an upper end portion of the earthquake-resistant connecting sleeve **212**, a second upper hook flange **214** provided at an upper end portion of the second upper long bolt **142B** and hooked on an inner surface of a lower end portion of the earthquake-resistant connecting sleeve **212**, and an earthquake-resistant spring **216** built into the earthquake-resistant connecting sleeve **212** and having upper and lower end portions connected to the first upper hook flange **213** of the first upper long bolt **142A** and the second upper hook flange **214** of the second upper long bolt **142B**.

(45) Therefore, when an earthquake occurs, the first upper long bolt **142A** and the second upper

long bolt **142B** are raised relative to the earthquake-resistant connecting sleeve **212** and the earthquake-resistant spring **216** absorbs vibrations due to the earthquake, and thus, the ceiling panel hanger **180** and the ceiling panel **120**, which are the main parts installed below the building slab, may be more reliably prevented from being easily shaken by vibrations and being damaged or detached due to the earthquake.

(46) Specific embodiments of the present disclosure have been described. However, the spirit and scope of the present disclosure are not limited to these specific embodiments, and those skilled in the art will understand that various modifications and variations may be made within the scope in which the gist of the present disclosure does not change.

(47) Therefore, these embodiments are provided only to completely disclose the present disclosure and for a person having ordinary skill in the art to which the present disclosure pertains, and therefore, they should be understood to be exemplary and not limiting in all aspects, and the present disclosure is defined only by the scope of the claims.

Claims

1. A ceiling panel mounting support structure with an improved mounting structure, the ceiling panel mounting support structure comprising: an E-shaped bracket (**130**) installed below a slab of a building; a turnbuckle (**160**) coupled to the E-shaped bracket (**130**); a ceiling panel hanger (**180**) including a pair of mounting portions (**182**) on both sides thereof based on a center and coupled to the turnbuckle (**160**), wherein ceiling panels (**120**) arranged to be adjacent to each other are mounted on the pair of mounting portion (**182**) of the ceiling panel hanger (**180**), wherein the E-shaped bracket (**130**) includes a vertical connecting bracket piece (**132**), an upper bracket piece (**134**) with a base portion coupled to one surface of the connecting bracket piece (**132**), a middle bracket piece (**136**), and a lower bracket piece (**138**) and having an E-shaped cross-section, the upper bracket piece (**134**) of the E-shaped bracket (**130**) is disposed on an upper surface of a lower beam piece (**116**) of an H-beam (**110**) installed below the building slab, the middle bracket piece (**136**) is fixed by an upper fixing bolt (**136UFB**) so that the E-shaped bracket (**130**) is fixedly installed to the H-beam (**110**), a long bolt (**140**) is coupled to the lower bracket piece (**138**) of the E-shaped bracket (**130**), and an upper connecting nut (**146**) and a lower connecting nut (**148**) coupled to a threaded portion of an outer circumferential surface of the long bolt (**140**) are respectively arranged on upper and lower surfaces of the lower bracket piece (**138**) of the E-shaped bracket (**130**) so that the long bolt (**140**) is fixed to the lower bracket piece (**138**) of the E-shaped bracket (**130**), the turnbuckle (**160**) has a ring shape with a pair of left and right side buckle pieces (**166**) on an upper buckle piece (**162**) and a lower buckle piece (**164**), and the threaded portion of the outer circumferential surface of the long bolt (**140**) is coupled to a buckle-side bolt (**172**) hole formed in the upper buckle piece (**162**) so that the turnbuckle (**160**) is coupled to the long bolt (**140**), an I-bolt (**170**) is coupled to the lower buckle piece (**164**) of the turnbuckle (**160**), and the ceiling panel hanger (**180**) is coupled to the I-bolt (**170**) by the bolt (**172**) and a nut (**174**); and a connecting reinforcement bracket (**190**), wherein the connecting reinforcement bracket (**190**) includes a reinforcement connecting piece (**192**) connected to the lower bracket piece (**138**) of the E-shaped bracket (**130**) via a hinge (HG), a cross reinforcement connecting piece (**193**) connected in a direction perpendicular to the reinforcement connecting piece (**192**), an upper reinforcement connecting piece (**194**) connected to an inner surface of the cross reinforcement connecting piece (**193**) and disposed in a direction parallel to the reinforcement connecting piece (**192**), a lower reinforcement connecting piece (**196**) connected to the inner surface of the cross reinforcement connecting piece (**193**) so as to be disposed below the upper reinforcement connecting piece (**194**) and extending in a direction parallel to the upper reinforcement connecting piece (**194**), and a reinforcement connecting support bolt (**198**) coupled to the lower reinforcement connecting piece (**196**) and having an upper end portion configured to protrude from a first side of the lower

reinforcement connecting piece (196) to a second side thereof.

2. The ceiling panel mounting support structure of claim 1, wherein the long bolt (140) includes an upper long bolt (142) having an upper end coupled to the lower bracket piece (138) of the E-shaped bracket (130) and a lower end to which a pipe hanger (150) is coupled and a lower long bolt (144) coupled to a lower end of the pipe hanger (150), and an upper connecting nut (146) and a lower connecting nut (148) fastened to a threaded portion of an outer circumferential surface of the lower long bolt (144) are respectively arranged on upper and lower surfaces of the lower bracket piece (138) of the E-shaped bracket (130) so that the long bolt (140) is fixed to the lower bracket piece (138) of the E-shaped bracket (130), the buckle-side bolt (172) hole formed in the upper buckle piece (162) of the turnbuckle (160) is coupled to a threaded portion of an outer circumferential surface of the lower long bolt (144) so that the turnbuckle (160) is coupled to the lower long bolt (144), and the ceiling panels (120) are installed by mounting the ceiling panels (120) adjacent to each other on a pair of the mounting portions (182) of the ceiling panel hanger (180).
